

Research Background

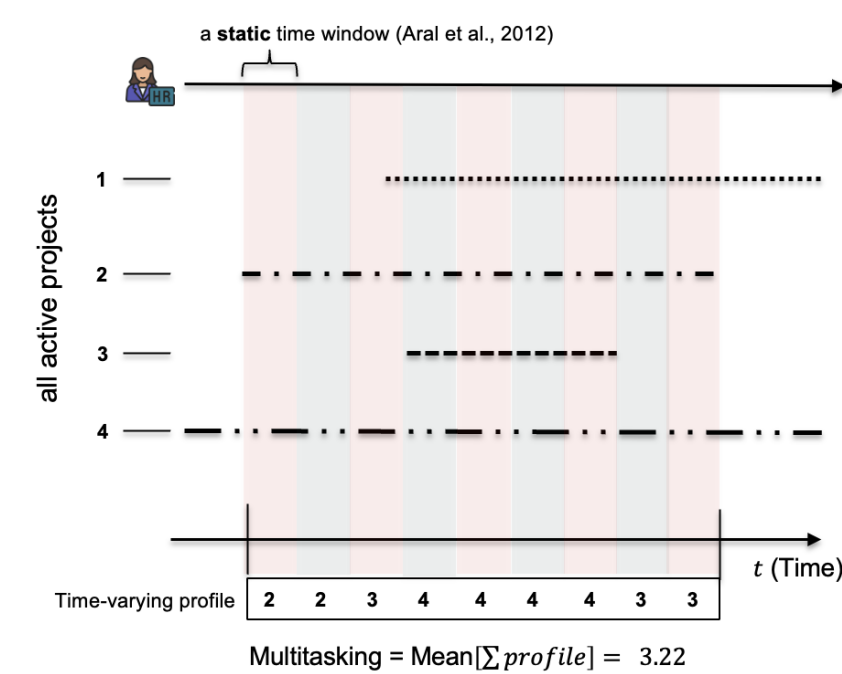
Extant understanding of multitasking

Contemporary organizational work is structured around multitasking, where employees handle multiple tasks simultaneously (i.e., simultaneous execution) and switch between multiple tasks sequentially (i.e., sequential switching) (Goes et al. 2018, Aral et al. 2012).

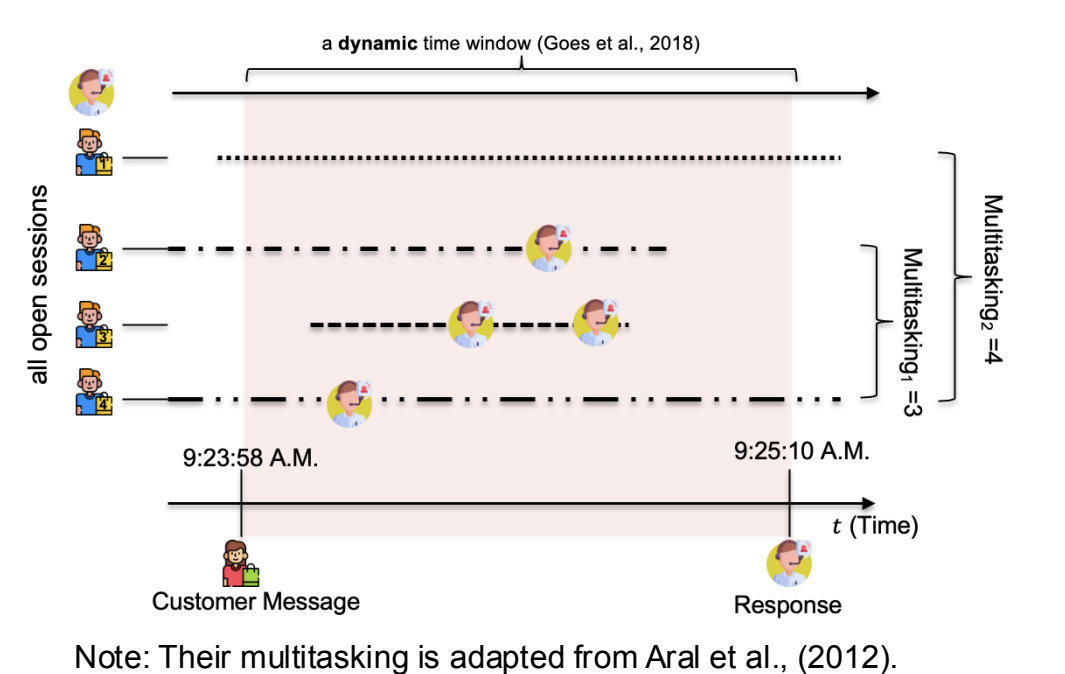
Traditionally, *multitasking* has been conceptualized as an individual-level phenomenon: *Multitasking is essentially a time-varying behavioral profile that reflects how an individual manages multiple tasks over time.*

Examples of individual multitasking as a time-varying profile

Example 1. One employee, Multiple recruiting projects (Aral et al. 2012)
Context: Recruiting



Example 2. One agent, Multiple customers (Goes et al., 2018)
Context: Live-chat



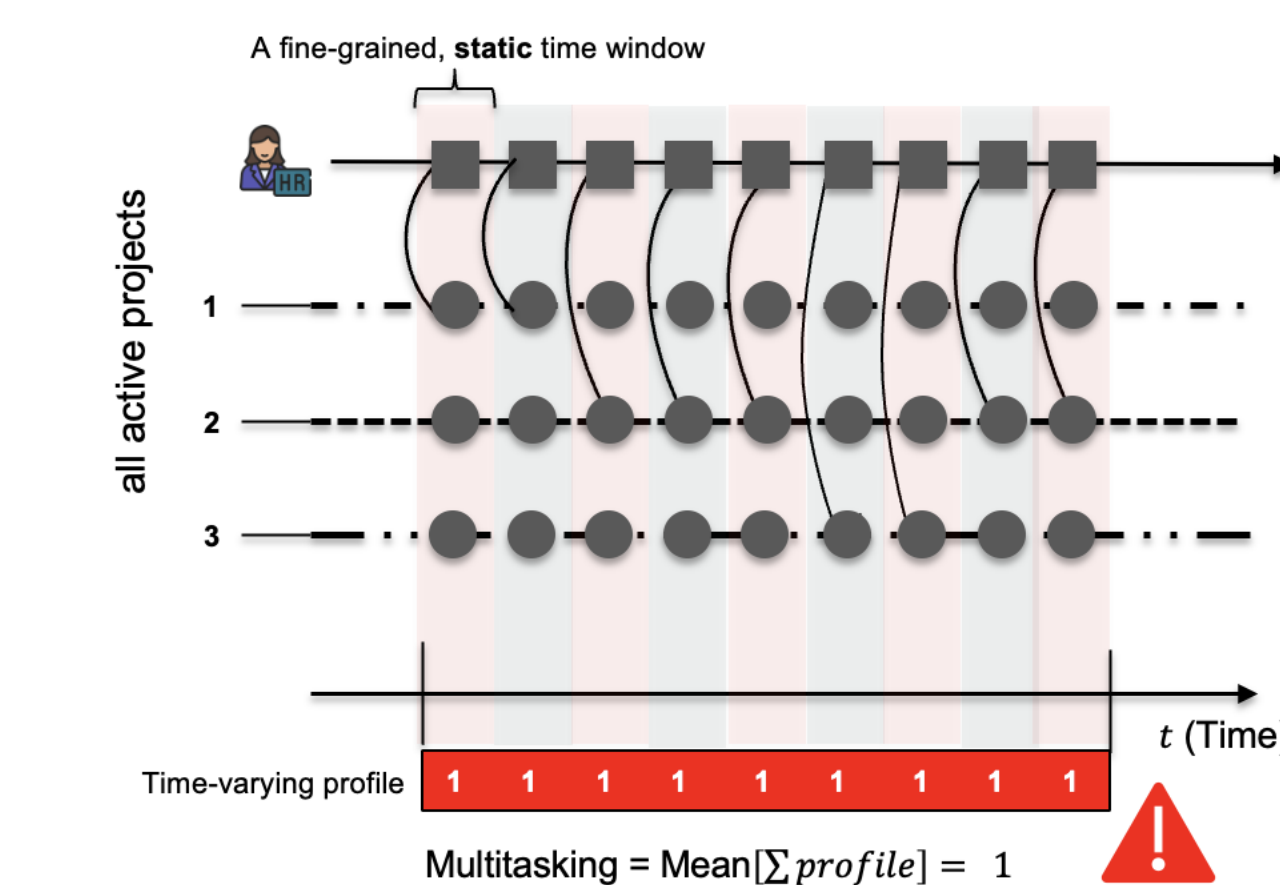
Problem

In practice, multitasking is more than an individual phenomenon. Multitasking is also more than a human phenomenon.

In organizations, teams perform multiple tasks ongoingly and at here-and-now, and multiple actors – humans and nonhumans – constitutively perform the multitasking phenomenon.

Why rethinking multitasking matters?

- Multitasking is the modern norm of work and is known to **affect work productivity**.
- Understanding multitasking can motivate managers to **mitigate the adverse effect of multitasking** on organizational outcome, including but not limited to worker productivity in terms of time and output, worker burdens and job satisfaction, and other performance measure such as customer satisfaction.
- Current understanding of multitasking from an individual worker's perspective does **not generalize** to more complex organizational work context.
- Current individualized view of multitasking is **criticized by researchers as a minomer** (Batt and Terwiesch, 2017)!



Criticizations

“Absolute parallel performance might not be possible as humans are unable to simultaneously divide their attention among ongoing tasks.”

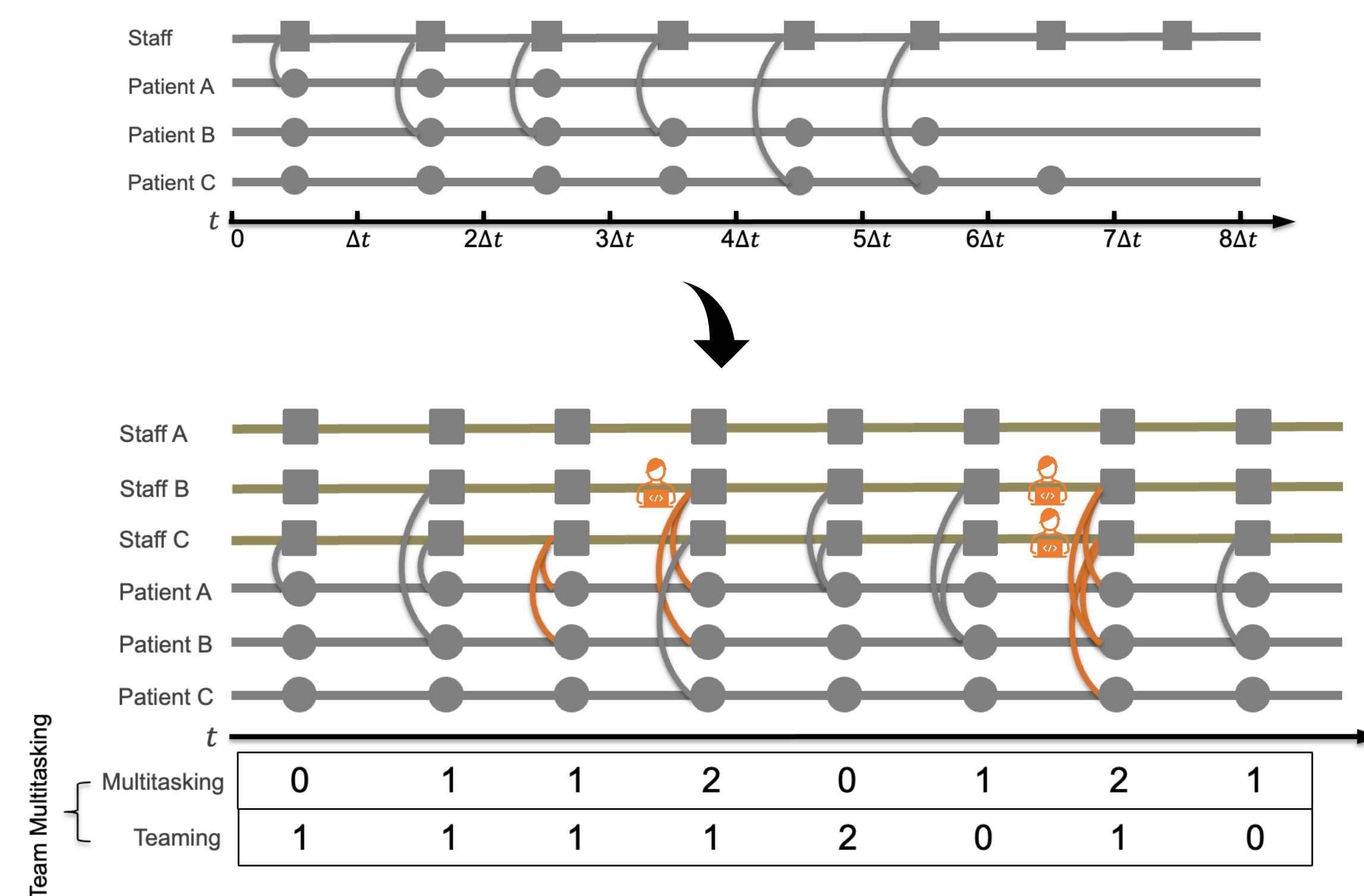
-- Benbunan-Fich et al. (2011, p. 3).

“Multitasking ... is not possible except when behaviors become completely automatic”

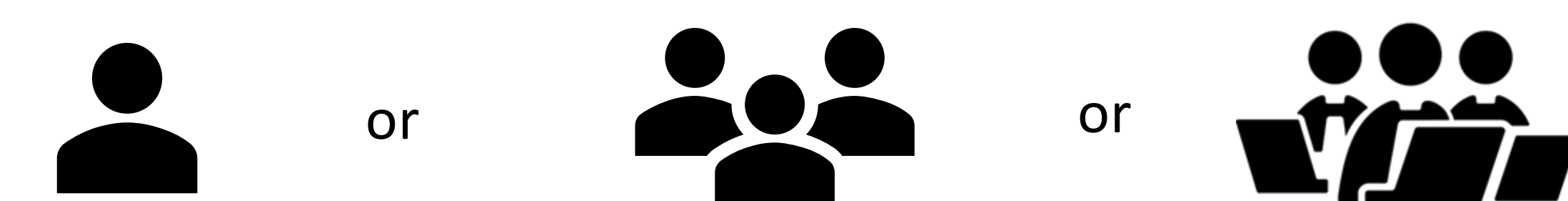
-- Skaugset et al. (2016).

Filling a Gap

Fill a research gap: Team multitasking



This work generalizes *multitasking as a time-varying behavioral profile that reflects how a **work unit** manages multiple tasks over time.* For a work unit, it can be an individual, a team, or a group of actors – human or nonhuman.



In practice, teaming is also a time-varying behavioral profile, which reflects the performative perspective of a team (see the figure above).

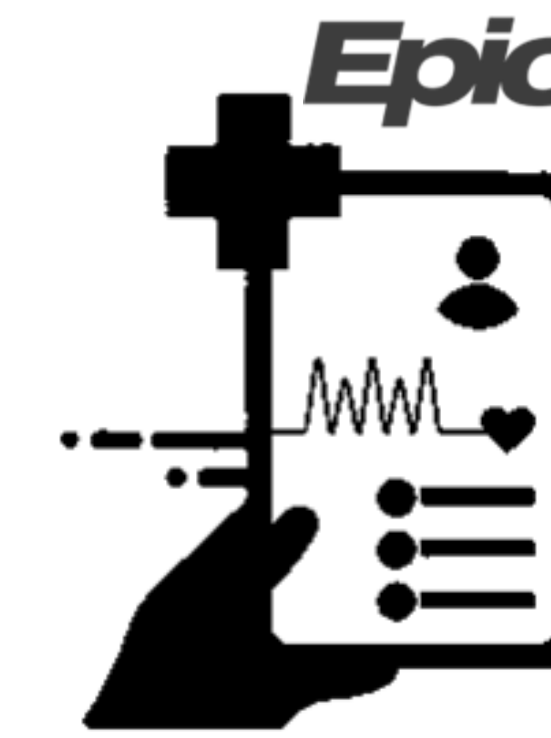
The gains with generalization of the concept

- Leaping from individual to team, multitasking should no longer be considered a misnomer. True multitasking exists as for every moment in time, multiple tasks can make progress simultaneously. By simultaneous progress, I mean, if you choose a time window Δt , then for every moment of time during the time window, multiple tasks are making progress. Such reconceptualization opens doors to **understand temporal simultaneity perspective of multitasking**, beyond the task switching perspective of multitasking.
- Leaping from human to actors (human and nonhumans), this work **foreground IT's role in multitasking**. Information systems do not just log the digital trace of human multitasking behaviors, but they also shadow how workers have appropriated IT features for multitasking including bulk processing. Furthermore, they shadow how IT sometimes auto-refreshes itself, leaving a digital footprint. For example, in the above figure, the orange edges show how one individual worker can bulk process multiple patient visits utilizing the IT features embedded in the healthcare information systems – EHR system, a.k.a. Electronic Healthcare Record system.

Research Context

Research Context

This work focuses on the **healthcare** context, especially the **outpatient clinics**, as it exemplifies the typical complex work organizing characterizing teamwork and multitasking, where individual conceptualization of multitasking does not apply, and in the meanwhile, where information systems are intensively used by healthcare workers, leaving **digital footprints** of their work behaviors.



Data

Sample EHR Audit Logs: one worker, multiple patient visits

Worker ID	Worker Role	Workstation ID	Action	Patient Visit ID	Timestamp	Diagnoses
Phy728	Physician	PLCCXR04	VISIT_DIAGNOSES	VIS143119	2020-09-02 12:00:55	Uncertain Neoplasm
Phy728	Physician	PLCCXR04	MR_REPORTS	VIS143119	2020-09-02 12:01:51	Uncertain Neoplasm
Phy728	Physician	PLCCXR04	UCNNOTE_PEND	VIS143206	2020-09-02 12:01:51	Lymphoma
Phy728	Physician	PLCCXR04	MR_RESULTS_EXIT	VIS143119	2020-09-02 12:01:55	Uncertain Neoplasm
Phy728	Physician	PLCCXR04	MR_RESULTS_EXIT	VIS143206	2020-09-02 12:01:55	Lymphoma
Phy728	Physician	PLCCDT37	MR_REPORTS	VIS143119	2020-09-02 12:08:17	Uncertain Neoplasm
Phy728	Physician	PLCCDT37	AC_VISIT_NAVIGATOR	VIS157805	2020-09-02 12:08:24	Lung Carcinoma
Phy728	Physician	PLCCDT37	AC_STORYLINE_VIEWED	VIS157805	2020-09-02 12:08:25	Lung Carcinoma
Phy728	Physician	PLCCDT37	MR_BPA_DISPLAYED	VIS157805	2020-09-02 12:08:29	Lung Carcinoma
...	...	...	...	...	...	...

Practice view: multiple team workers, multiple patient visits



Measures

Independent Variable	
Team multitasking	The number of patient visits served by a focal team, adapted from Aral et al. (2012). If performing patient visit level of analysis, the focal patient visit is excluded when counting the number of patient visits.
Dependent Variable	
EHR documentation time	The total E-documentation time of a team. If performing patient visit level of analysis, the E-documentation time is the sum of documentation time of all team members of the focal team.
Clicks	The total number of clicks of a team.
Time per Click	The time spent on each click, suggesting the lags of clicks on average.
Key Control Variables	
Teaming	The average teaming level (on focal patient visit) per unit of time.
Team size	The number of team members serving a patient visit, following Aral et al. (2012).
Diagnosis prevalence	Adapted from existing literature that controls for task heterogeneity. Excluding this control when using FE model.
Task duration	Follow the existing literature that controls for task duration when investigating the multitasking impact.
Timing	Time dummy including morning, noon, and afternoon. This follows existing literature in controlling for timing when investigating the multitasking impact, for example, the contact time when multi-chatting with multiple customers online.
IT-related variables	
IT multitasking	<i>hasITmultitasking</i> : A binary variable indicating whether any team member in a team enact IT multitasking during a patient visit.
Frequency of IT multitasking	<i>nITmultitaskingActions</i> : The frequency that IT-multitasking is enacted by a team during a focal patient visit.
Number of team members enacting IT multitasking	<i>nWorkersUsingITmultitasking</i> : The number of team workers enacting IT-multitasking during a focal patient visit.

Key Findings and Takeaways

Empirical Model

Patient visit level analyses:

$$Y_{vdt} = TeamMultitasking_{vdt} + Controls_{vdt} + \alpha_d + \tau_t + \varepsilon_{vdt}$$

, where we include diagnosis FE and week FE.

Key findings

	Dependent Variable							
	LogEHRDocumentationTime				nClicks			
TeamMultitasking	1.820*** (0.180)	1.820*** (0.196)	1.840*** (0.189)	295.9*** (11.8)	284.7*** (12.6)	287.9*** (12.1)	3.15*** (0.276)	3.35*** (0.280)
TeamMultitasking ²	-0.812*** (0.069)	-0.810*** (0.068)	-0.803*** (0.066)	-75.1*** (5.51)	-75.9*** (5.42)	-74.7*** (5.40)	-1.78*** (0.147)	-1.77*** (0.148)
Teaming	1.970*** (0.174)	1.960*** (0.179)	1.970*** (0.175)	335.6*** (10.9)	327.4*** (11.1)	329.4*** (11.0)	4.19*** (0.202)	4.31*** (0.198)
hasITmultitasking	0.196* (0.107)			4.36 (7.33)			0.485 (0.295)	
nITmultitaskingActions		0.089** (0.025)			17.100*** (1.51)		-0.030 (0.047)	
nWorkersUsingITmultitasking			0.062* (0.036)			11.600** (3.330)	-0.058 (0.069)	
Controls	Yes							
Diagnosis FE	Yes							
Week FE	Yes							
S.E. Clustered	by: diagnosis & week							
Observations	3,684	3,684	3,684	3,684	3,684	3,684	3,684	3,684
Within R2	0.585	0.585	0.585	0.760	0.763	0.761	0.182	0.181
Note:	p<0.1; **p<0.05; ***p<0.01							

The variables (excluding the binary variables) are standardized for improved interpretation and across variable comparisons.

Main results of this study show the followings.

- **Team multitasking** has a **positive but diminishing effect** on healthcare team's E-documentation burden (see the first and second row).
- Compare to team multitasking, **IT multitasking** slightly **increases** healthcare team's E-documentation time and overall number of clicks but does not increase the time lag of each click, suggesting using IT multitasking does not increase team workers' cognitive burden.

Takeaways

- Lens of practice (Feldman and Orlikowski., 2011): Practice lens or practice theory focuses on **dynamics, relations, and enactment**. Through this work, we illustrates how **digital traces or footprints opens possibilities of adopting such lens** in improving our understanding the complex work practice.
- Path theory (Pentland et al., 2025): Multitasking can connect to path theory, a meta theory. As this work illustrates, multitasking is essentially **the number of unique (projects) paths a work unit intersects with** during a time window, be it a static or a dynamic time window. The intersections are time-varying, and these time-varying intersections have an empirical impact on organizational outcomes, illustrating the applicability of path theory on answering practical questions.
- Sociomateriality (Orlikowski., 2007): EHR system becomes pervasive in healthcare organizations. Although multitasking is not a sociomaterial concept, **IT-multitasking in this work can be considered as a sociomaterial concept**, as IT-multitasking exists in a sociomaterial way. We cannot talk about IT-multitasking without technology nor without humans' enactment or initiatives.

Thank you for your attention
Suggestions are welcome!